

SUDHESHNA REDDY THIRUMALAREDDY

☎ +91-9182187075 ✉ sudheshnareddy.tr@gmail.com  [LinkedIn](#)  [GitHub](#)  [LeetCode](#)  [Codolio](#)

Education

Indian Institute of Information Technology and Management, Gwalior

Nov 2022 – Present

Integrated B.Tech and M.Tech in Information Technology, CGPA: 8.04

(Expected Graduation: May 2027)

Technical Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL

Frameworks & Tools: HTML, Tailwind CSS, React.js, Next.js, Node.js, Express.js, Postman, RESTful APIs, FastAPI

Databases: MongoDB, PostgreSQL, MySQL, Firebase

Tools & Platforms: Git, GitHub, GitHub Actions, Docker, VS Code, AWS (S3, EC2, RDS, Lambda), Vercel, Kafka

Relevant Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Management Systems, Agile Development, Operating Systems, Computer Networks, Machine Learning, System Design

Experience

Zetwerk

May 2025 – Jul 2025

Software Engineer Intern

Bangalore

- Architected an **AI-powered Analytics Sandbox** for bank statement PDFs, supporting **20+ standard attributes**, custom AI-generated attributes, multi-run experiments, output comparison, and LLM-based result querying.
- Designed a **6-step backend workflow** covering S3 presigned PDF uploads, sandbox creation, run creation, asynchronous analytics execution, status polling, and metadata persistence across **RDS and DynamoDB**.
- Engineered financial analytics pipelines across **4 modules**: portfolio metrics, transaction insights, benchmark comparisons, and monthly returns, supporting scheduled and on-demand data refreshes.

Projects

ParkWise Bengaluru Intelligence  | **Live Demo**  | *React, TypeScript, PostgreSQL, Docker, LightGBM* Jun 2026

- Built an **end-to-end parking congestion intelligence platform** for Flipkart Gridlock 2.0 Round 2, covering data ingestion, hotspot detection, forecasting, officer allocation, and patrol routing.
- Processed **298K+ Bengaluru traffic violation records** using PostgreSQL/PostGIS and DBSCAN-based **geospatial clustering** to detect parking-induced congestion hotspots.
- Developed a **Dockerized FastAPI backend** and **React dashboard**, using **LightGBM** to forecast future hotspot risk with **0.87 R²** and **3.30 MAE**.

NextFlow  | **Live Demo**  | *Next.js, TypeScript, React Flow, Prisma, PostgreSQL, Trigger.dev* May 2026 – Jun 2026

- Built **NextFlow**, a full-stack AI workflow automation platform with **6+ node types**, enabling users to create, execute, and persist multi-branch workflows using **Next.js, React Flow, Prisma, PostgreSQL, and Zustand**.
- Designed a **DAG-based execution engine** using topological sorting to process **10+ dependent nodes** in order, execute independent branches in parallel, and orchestrate background runs via **Trigger.dev**.
- Integrated **3 external services**: **OpenRouter/Gemini**, **FFmpeg**, and **Transloadit** for LLM, media processing.

CareerCrawler - AI Job Finder  | **Live Demo**  | *React, Node.js, Express.js, Web Scraping* Mar 2026 – Apr 2026

- Architected a job aggregation platform centralizing **500+ live listings** from multiple career portals into indexed **MongoDB** collections, reducing candidate search time across **10+ sites**.
- Developed multi-site scraping pipelines using **Axios, Cheerio, and Puppeteer** to extract job data from **Greenhouse, Lever, and Workday**, achieving **98% extraction accuracy**.
- Deployed automated scrapers and **REST APIs**, linking a **React/Vercel** frontend with a **Node.js/Render** backend.

Publications

EchoShield: Understanding Structural Hole Spanners to Mitigate Echo Chambers  May 2026

Published at 18th ACM Web Science Conference Companion Volume

Germany

Achievements

- Achieved **Rank 33** in Codeforces Round 1101, **Rank 122** in CodeChef Starters 229, and **Global Rank 655** in LeetCode Weekly Contest 503.
- Solved **900+ DSA problems** and earned **Knight** on LeetCode, **Expert** on Codeforces, and **3-star** on CodeChef.
- Achieved **Global Rank 1358** in Meta Hacker Cup 2025 Round 2 among **20,000+ participants**.
- Advanced to **Round 2** of Flipkart Gridlock 2.0 Hackathon, ranking among the **top 1,630 teams** out of **30,000**.
- Ranked **1372** in the Goldman Sachs 2025 Hackathon among **15,975 teams**.
- Secured **1st Rank** at IIITM Gwalior in the aptitude test conducted by Naukri EROH'25.